

Fruit Battery

Experiment Packet

Audience: instructors and teaching assistants

Purpose: Bench instructions for instructors and TAs

Session hook: Familiar objects make electricity

Length: 90-minute lab block (~20 min concepts + ~70 min hands-on)

This packet includes

1. Session overview & plan
2. Preparation
3. Materials checklist
4. Experiment procedure
5. Shared safety notes

Generated from the Electrochemistry workshop curriculum. Prefer the latest PDF from the course repository if procedures change mid-week.

Session 1 — Build a Fruit Battery

Hook: Familiar objects make electricity.

Learning outcomes

By the end of this session, students should be able to:

- Build a simple electrochemical cell from fruit and metal electrodes
- Explain oxidation and reduction in a galvanic cell
- Identify anode and cathode and describe electron vs. ion flow
- Measure voltage with a multimeter
- Explain why cells in series add voltage
- Describe why batteries eventually die (internal resistance, reactant depletion)

Session plan (90 min)

Time	Activity	Notes
0–10 min	Hook: measure one lemon cell; ask whether a lemon can make electricity	See lecture.md — Opening hook
10–25 min	Concept: zinc oxidation, copper cathode, acid electrolyte, electron flow	See lecture.md
25–45 min	Build one cell; measure voltage	See experiment.md — Part 1
45–60 min	Series battery challenge	See experiment.md — Part 2
60–75 min	Power LED, buzzer, or clock; discuss V, I, and internal resistance	See experiment.md — Part 3
75–90 min	Side experiments: fruits, metals, pH, salt, parallel; wrap-up + preview Session 2	See experiment.md — Extensions

Key reactions

Anode: $\text{Zn(s)} \rightarrow \text{Zn}^{2+}(\text{aq}) + 2\text{e}^{-}$

Cathode: $2\text{H}^{+}(\text{aq}) + 2\text{e}^{-} \rightarrow \text{H}_2(\text{g})$

Overall: $\text{Zn(s)} + 2\text{H}^{+}(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{H}_2(\text{g})$

Files in this folder

File	Purpose
lecture.md	~20 min concept block

File	Purpose
experiment.md	Hands-on procedures and data
materials.md	Session checklist
preparation.md	Pre-class setup

Instructor reminders

- Roll lemons/limes first to break internal membranes
- Cells in series add voltage ~linearly; current stays limited
- Galvanized nail = steel + zinc coating (zinc is the active metal)
- pH near cathode may become less acidic (protons consumed)

Status

- ☐ Lecture outline finalized
- ☐ Experiment procedure tested
- ☐ Materials procured
- ☐ Session 2 CuSO₄ crystals on hand for next day

Session 1 — Preparation

Before class (instructor)

- ☐ Procure fresh fruit (firm, not dried out)
- ☐ Test one cell: expected ~0.8–1.0 V for lemon + Zn + Cu
- ☐ Verify multimeters have fresh batteries and work on DC voltage
- ☐ Pre-cut copper strips if students struggle with stiff wire
- ☐ Prepare waste collection for fruit and electrodes

Day-of setup (15 min before)

- ☐ One station per group: fruit, electrodes, wires, multimeter, load device
- ☐ Optional extension materials at a side table

After class

- ☐ Dispose of fruit waste appropriately
- ☐ Rinse and dry electrodes if reusing
- ☐ Confirm Session 2 $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is on site for tomorrow's plating bath

Open questions / notes

- Typical voltage achieved with our materials:
- Best fruit type in pilot:
- LED lighting requires ____ cells in series:

Session 1 — Materials

Per group (3–4 students)

- ☐ 3–6 lemons, limes, potatoes, apples, or oranges
- ☐ Galvanized nails or zinc strips (2+)
- ☐ Copper strips, copper wire, or copper coins/pennies (2+)
- ☐ Alligator clip wires (4–6)
- ☐ Multimeter
- ☐ LED, small buzzer, or low-power digital clock
- ☐ pH paper or pH indicator strips
- ☐ Paper towels

Optional (extensions)

- ☐ Graphite pencil leads
- ☐ Aluminum foil
- ☐ Steel nails
- ☐ Magnesium ribbon
- ☐ Table salt (small amount for salt-addition test)

Instructor / room

- ☐ Waste bin for used fruit (not food waste)
- ☐ Hand-washing access
- ☐ Demo multimeter on projector or large display (optional)

Quantities to estimate

Item	Groups	Total
Fruit (minimum)	× 4 each	
Multimeters	1 per group	
9 V batteries	—	Not needed this session

Session 1 — Experiment: Fruit Battery

Main experiment: Lemon/lime/potato battery — generating electricity from chemistry.

Instructors: expected voltages, LED failure modes, and extension interpretation — instructor.md.

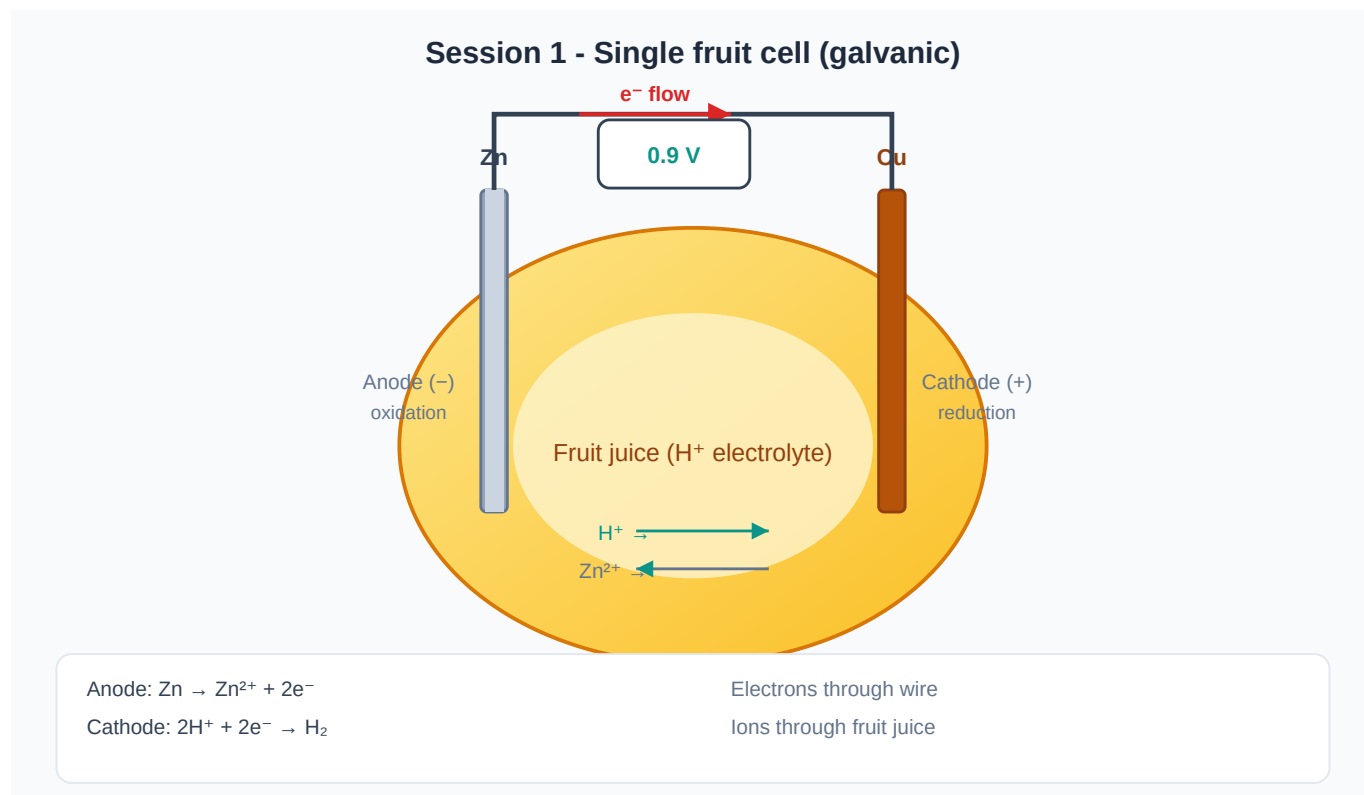


Figure 1 — Single galvanic cell: Zn anode (-), Cu cathode (+), electrons through the wire, ions through the fruit juice.

Materials checklist (per group)

Item	Spec / notes
Fruit	3–6 firm lemons, limes, potatoes, or apples
Zinc electrode	Galvanized nail or zinc strip (2+)
Copper electrode	Copper strip, stiff wire, or clean penny (2+)
Multimeter	DC voltage (and current if available)
Alligator clips	4–6 leads
Load	Red LED (1.8–2.2 V forward) and/or 1.5–3 V buzzer
Optional	pH paper, table salt, aluminum foil

Part 1 — Build one cell (25–45 min)

Procedure

1. Choose one fruit. **Roll** a citrus fruit firmly on the bench for 10–15 s (breaks juice sacs). For potato/apple, squeeze gently or make a shallow slit for better contact.
2. Insert the **zinc** electrode (galvanized nail or zinc strip) about halfway in — do not push all the way through.
3. Insert the **copper** electrode **2–3 cm away** from the zinc. Electrodes must **not touch** inside the fruit (touching shorts the cell → ~0 V).
4. Set the multimeter to **DC voltage** (V $\overline{=}$), typically the 2 V or 20 V range.
5. Clip **red (V Ω mA)** lead to **copper**; clip **black (COM)** to **zinc**.
6. Expected: copper reads **positive** relative to zinc (~0.8–1.0 V for lemon).
7. Record the open-circuit voltage. Wait 2 minutes; record again (note drift).

Expected results

Fruit (typical)	Open-circuit voltage
Lemon / lime	~0.8–1.0 V
Potato / apple	~0.5–0.9 V (varies)

Observations to record

Quantity	Value	Notes
Fruit type		
Electrode materials	Zn: ____ / Cu: ____	
Electrode separation (cm)		
Open-circuit voltage (V)		
Voltage after 2 min (V)		
Meter polarity (which metal +?)		

Troubleshooting

Problem	Likely cause	Fix
~0 V	Electrodes touching	Separate inside the fruit
Very low V	Dry fruit, poor contact	Roll fruit; re-insert; try fresher fruit
Unstable reading	Loose clips	Secure alligator clips on bare metal
Negative voltage	Leads swapped	Swap clips or note polarity and continue

Part 2 — Series battery challenge (45–60 min)

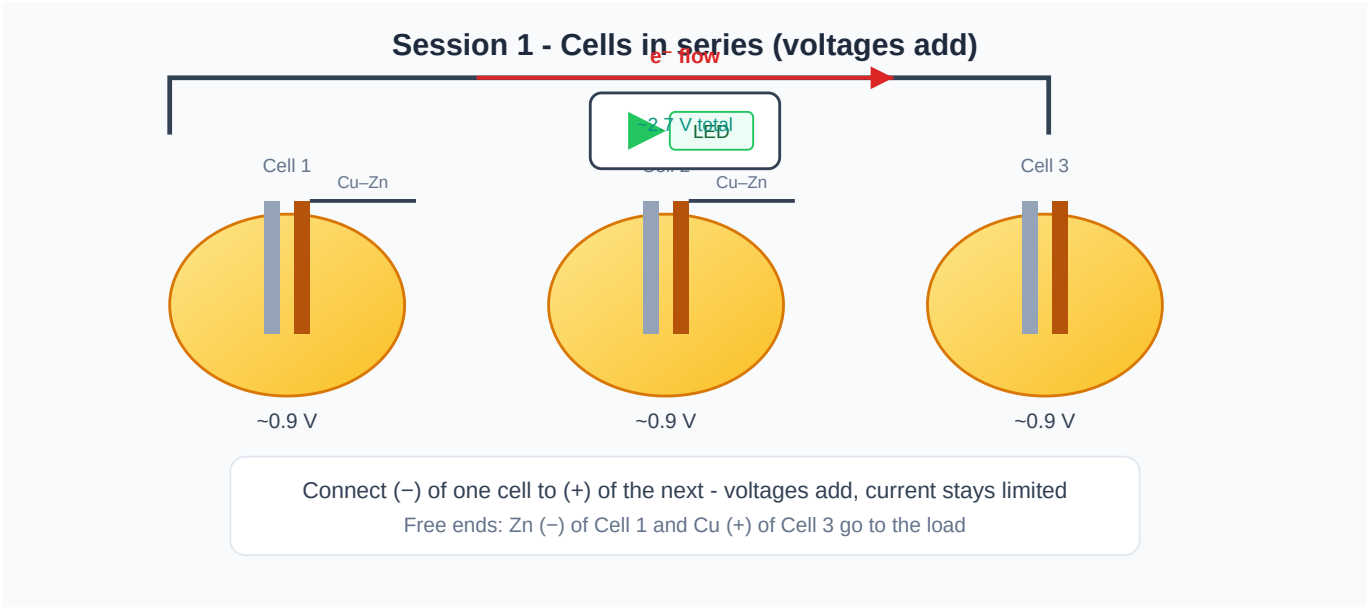


Figure 2 — Cells in series: connect Cu of one cell to Zn of the next; voltages add (~0.9 V each).

Procedure

- 1. Build 2–3 identical cells (one fruit each) the same way as Part 1.
- 2. Measure and record each cell's voltage alone first.
- 3. Wire **in series**:
- 4. Copper of cell 1 → zinc of cell 2 (alligator lead)
- 5. Copper of cell 2 → zinc of cell 3
- 6. Measure total voltage between the **free copper** (positive end) and **free zinc** (negative end).
- 7. Compare measured total to the sum of the individual voltages.

Challenge question

"If each cell is 0.9 V, how many cells do you need for a 3 V buzzer? For a 1.5 V clock?"

Data table

# cells in series	V ₁	V ₂	V ₃	V _{total} measured	V _{sum} individual	Match?
1		—	—			
2			—			
3						

Expected

3 lemon cells ≈ 2.4–3.0 V open circuit (enough in voltage for many red LEDs; current may still be limiting).

Part 3 — Power something (60–75 min)

Load options (try in order)

1. Multimeter on **DC mA** current range (small load — should work)
2. **Red LED** — longer leg / flat side conventions: longer leg usually + (to copper end of series string). Needs ~2 V+ and a few mA. Use **3 cells** first.
3. Active buzzer rated **1.5–3 V DC** (with leads)
4. Low-voltage potato/lemon clock kit (if available)

LED wiring reminder

- Series string: free **copper** → LED **anode (+)**; LED **cathode (-)** → free **zinc**
- If it does not light, reverse the LED once before concluding failure

Measurements

Load	# cells	Voltage under load (V)	Current (mA)	Works?
Open circuit			—	
LED				
Buzzer/clock				

Discussion after attempt

- Did voltage **drop** when connected to a load? (Internal resistance)
 - Why might the LED need more **current** than the fruit cell can supply?
-

Extensions — Side experiments (75–85 min)

Students choose one or more:

A. Compare fruits

Same Zn/Cu electrodes, different fruits → compare open-circuit voltage.

B. Compare metals

Same fruit; replace Zn with aluminum foil, steel nail, or Mg (if available) vs. copper. Record V and note any oxide-film issues (especially Al).

C. pH changes

After 10+ min under load, touch pH paper near each electrode. Hypothesis: cathode region may become less acidic.

D. Salt addition

Add a **pinch** (~0.25 tsp) of table salt into a slit in the fruit; re-measure V and current under the same load.

E. Parallel vs. series

Two cells in parallel (Cu–Cu and Zn–Zn tied): voltage $\approx$ one cell; compare current capability to series if meters allow.

Extension report template

Variable changed	Control	Result	Explanation (hypothesis)

Wrap-up (85–90 min)

1. Clean stations; fruit goes to waste (not food waste).
2. Quick share: best voltage, whether the LED lit, one surprising result.
3. Preview Session 2: *tomorrow we dissolve copper sulfate and use electricity to plate copper onto a nail.*

Student handout — Quick reference

Build: Roll fruit $\rightarrow$ Zn + Cu, not touching $\rightarrow$ measure V

Series: Cu of one cell to Zn of the next $\rightarrow$ voltages add

LED: Needs $\sim$ 3 cells + correct polarity; voltage may sag under load

Safety: Do not eat experimental fruit; wash hands after metals

Experiment status

- ☐ Procedure pilot-tested with actual fruit/electrodes
- ☐ Typical voltage range documented (record: ____ V)
- ☐ LED/buzzer compatibility verified (# cells needed: ____)
- ☐ Extension options ranked by time available
- ☐ Wrap-up / Session 2 preview timed

Syllabus Safety Notes

Apply these across all sessions. Session-specific hazards are listed in each session folder.

The three rules that matter most

1. **Salt belongs in exactly one place this week.** NaCl is used only for the Session 5 corrosion jars, where nothing is driven hard enough to make chlorine. It is **forbidden** in every electrolysis bath and in the derusting cells, where chloride is oxidized at the anode to **chlorine gas**. This is a competing half-reaction, not an arbitrary rule.
2. **Any cell left running unattended must be open and ventilated.** Electrolysis and derusting cells evolve hydrogen continuously. An overnight derusting cell can generate a couple of litres of it. Open jars, ventilated room, no ignition sources, nothing sealed. Ever.
3. **Disconnect power before touching anything wet.** Every time, as a ritual, especially at the Session 5 reveal.

General rules

1. **No eating experimental materials.** Fruit, potatoes and liquids used in experiments are no longer food, and this includes the ones that started out edible.
2. **Ammeters go in series.** A meter on a current range clipped across a supply is a short circuit and will blow its fuse at best.
3. **Avoid short circuits generally.** Batteries heat up fast when shorted.
4. **Collect metal-salt waste.** Copper and silver solutions are not poured down the drain.
5. **Water choice:** tap water is fine for most baths. **Distilled is required for the Session 3 silver bath**, so chloride does not precipitate silver, and is worth having for the first rung of the Session 4 conductivity ladder.
6. **Goggles for all wet work.** Gloves additionally for silver nitrate.
7. **Rusty hardware is sharp** and often outdoors-dirty. Forceps or gloves; no bare-handed scrubbing.

Session-specific hazards

Session	Key hazard	Mitigation
1	Sharp nails and wires	Supervise handling; wash hands after handling metals
2	Copper sulfate solution	Goggles; no ingestion; collect waste
3	Silver nitrate	Gloves and goggles; stains skin and clothing near-permanently; small volumes (30–40 mL); collect all waste; no ammonia, ever
4	Gas evolution; hydrogen ignition	No NaCl; ammeter in series; U-tube side arms open; balloon and flame instructor only
5	Corrosion demos; overnight derusting	Washing soda only, never NaCl; disconnect before handling; jars open and ventilated overnight

Silver nitrate and ammonia — Session 3

Never add ammonia, or any ammonia-containing cleaner, to a silver nitrate solution. Ammoniacal silver solutions (Tollens' reagent) deposit **silver nitride and related compounds** on standing, which are friction- and shock-sensitive explosives. The hazard is not the fresh solution — it is any residue left in a bottle, on a stirrer, or around a cap.

This matters because the "silver mirror" recipes students will find online use exactly this chemistry. Say so out loud during the Session 3 safety brief, before someone tries it at home.

It also does not even work for our purpose: ammonia complexes copper about as strongly as it complexes silver, so it does nothing to suppress the displacement reaction we are trying to control.

Hydrogen — Session 4, instructor only

The balloon demo is optional. The chemistry is fully demonstrated by the splint tests without any of this risk, so skip the balloon entirely if the venue forbids open flame or if you are at all unsure.

If you do run it:

- Feed **cathode (H₂) gas only** into the balloon. The anode vents separately to the room.
- **Never a deliberate H₂/O₂ mixture.** Pure hydrogen in air *deflagrates* — it burns at the surface and gives a bark. A stoichiometric mixture **detonates**, which is a different event entirely and not a classroom demonstration.
- **Never a rigid or sealed container.** A balloon tears; glass becomes shrapnel.
- Modest inflation only.
- Tie off, then carry it **well away** from the electrolysis cell, which is still making hydrogen.
- Long lighter at arm's length; area cleared; extinguisher present and its location known.
- Warn the room. The bang is loud, sudden, and some students find it distressing.

Splint tests — the low-risk alternative

Gas	Test	Result
H ₂	Lit splint at the mouth of the tube	Squeaky pop
O ₂	Glowing (not flaming) splint inserted	Ember relights

A few millilitres of gas in a small test tube is plenty. Keep hydrogen tubes **mouth-down** until the moment of testing. TA-supervised; students may hold the tube, an adult holds the flame.

Waste

Waste	Disposal
Copper sulfate solution	Labeled waste container; not down the drain
Silver nitrate solution and rinses	Labeled waste container; not down the drain. Drop a coil of bare copper wire in — it strips the silver back out as metal for recovery
Sodium sulfate electrolyte	Drain is acceptable; dilute well

Waste	Disposal
Washing soda derusting sludge	Let solids settle, bag the sludge, dilute the liquid to drain
Saltwater corrosion jars	Drain; dispose of rusty metal in solid waste